

Q-CANVAS

A Quantum Simulation Platform for
Multi-Framework Quantum Programming

Supervised By:

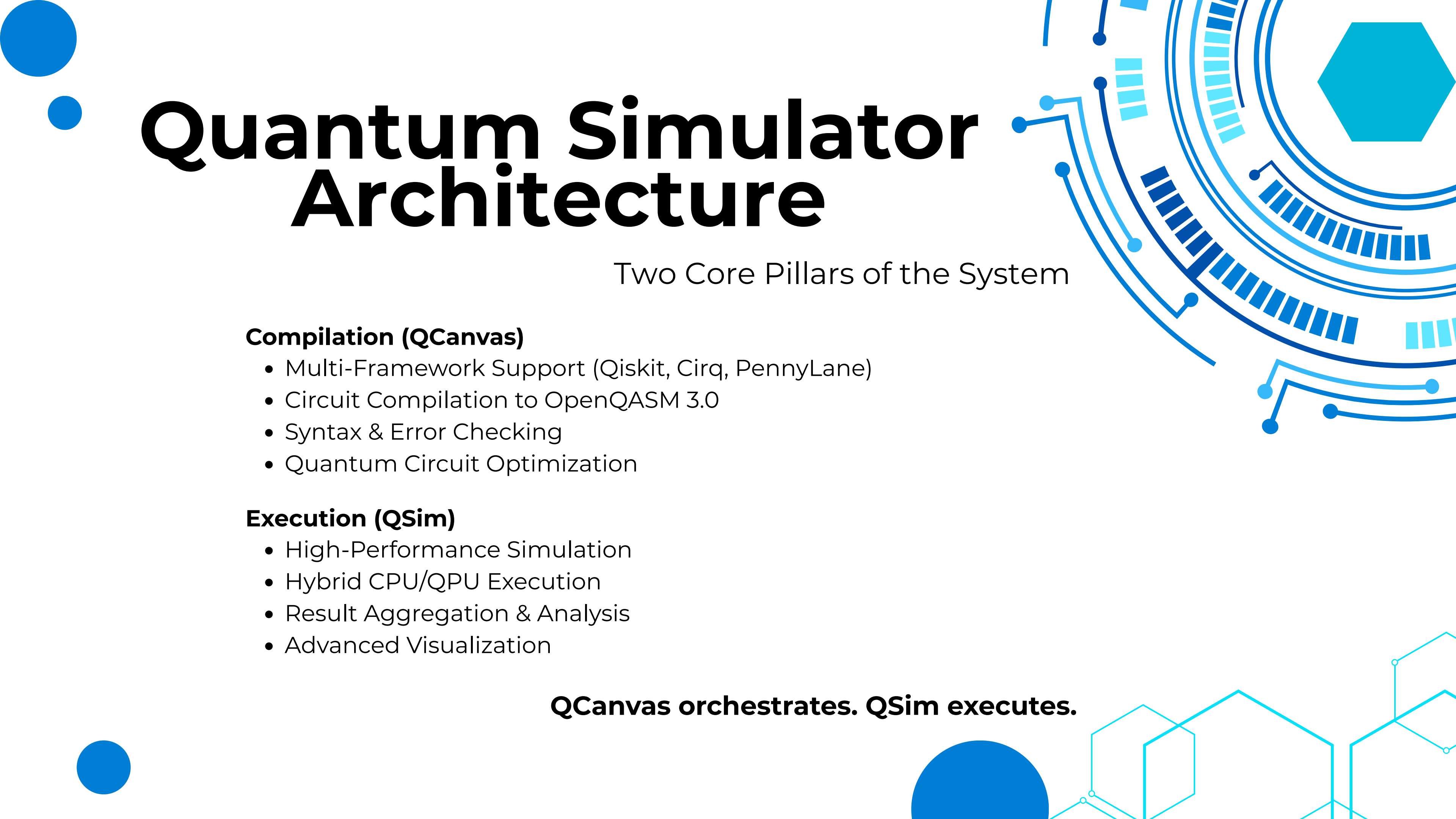
Dr. Imran Ashraf

Presented By:

Umer Farooq

Hussain Waseem Syed

Muhammad Irtaza Khan



Quantum Simulator Architecture

Two Core Pillars of the System

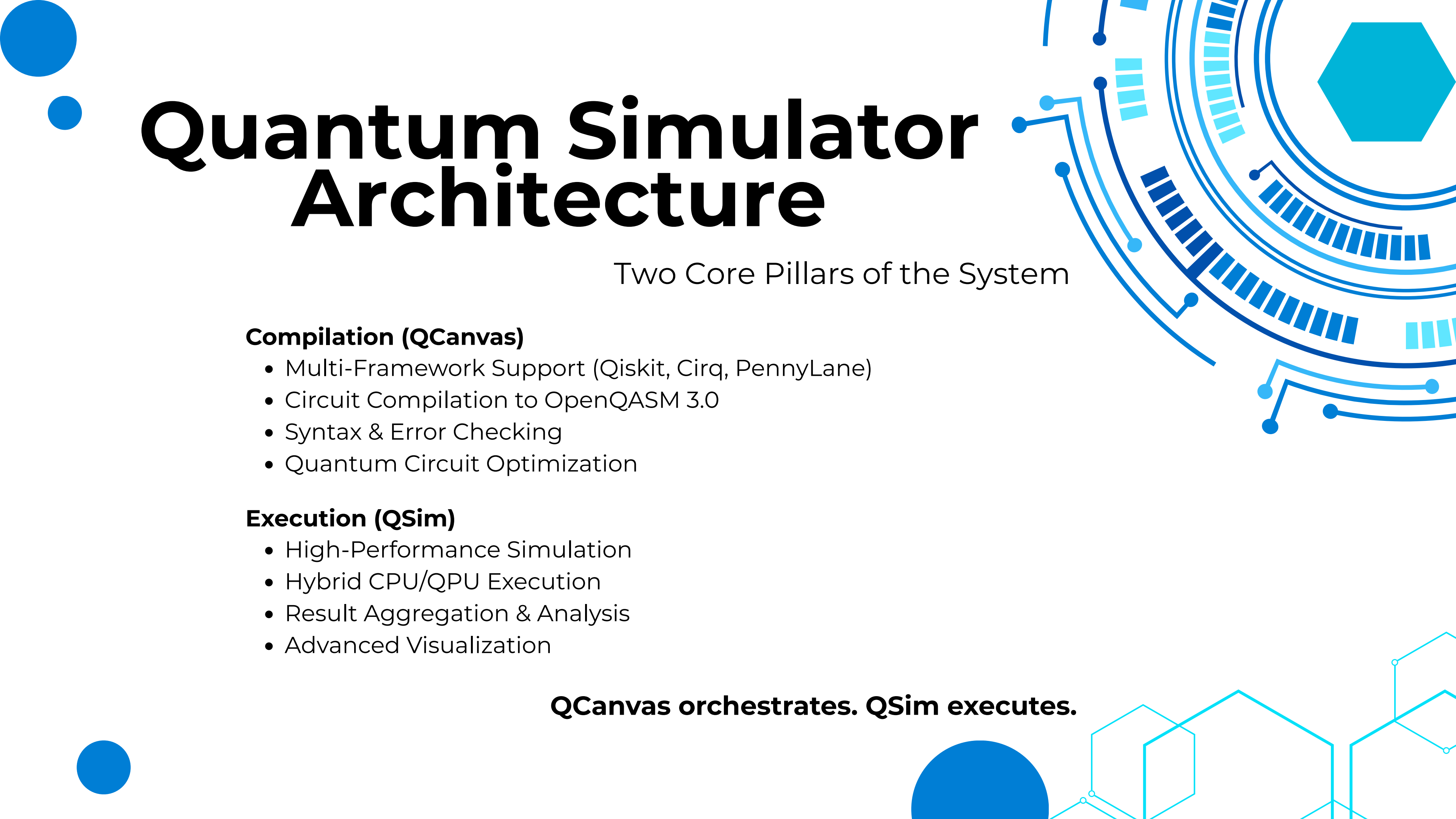
Compilation (QCanvas)

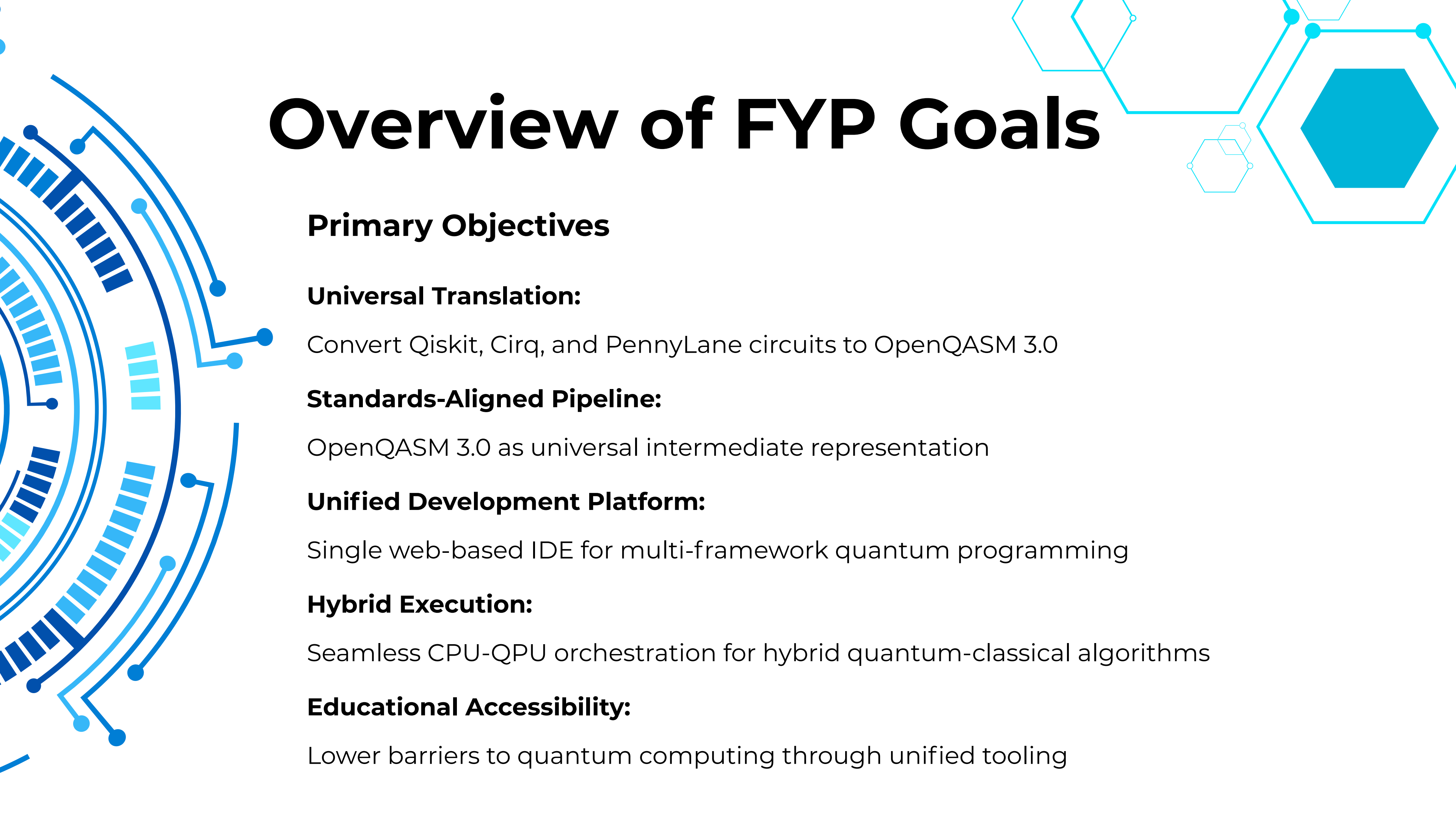
- Multi-Framework Support (Qiskit, Cirq, PennyLane)
- Circuit Compilation to OpenQASM 3.0
- Syntax & Error Checking
- Quantum Circuit Optimization

Execution (QSim)

- High-Performance Simulation
- Hybrid CPU/QPU Execution
- Result Aggregation & Analysis
- Advanced Visualization

QCanvas orchestrates. QSim executes.





Overview of FYP Goals

Primary Objectives

Universal Translation:

Convert Qiskit, Cirq, and PennyLane circuits to OpenQASM 3.0

Standards-Aligned Pipeline:

OpenQASM 3.0 as universal intermediate representation

Unified Development Platform:

Single web-based IDE for multi-framework quantum programming

Hybrid Execution:

Seamless CPU-QPU orchestration for hybrid quantum-classical algorithms

Educational Accessibility:

Lower barriers to quantum computing through unified tooling

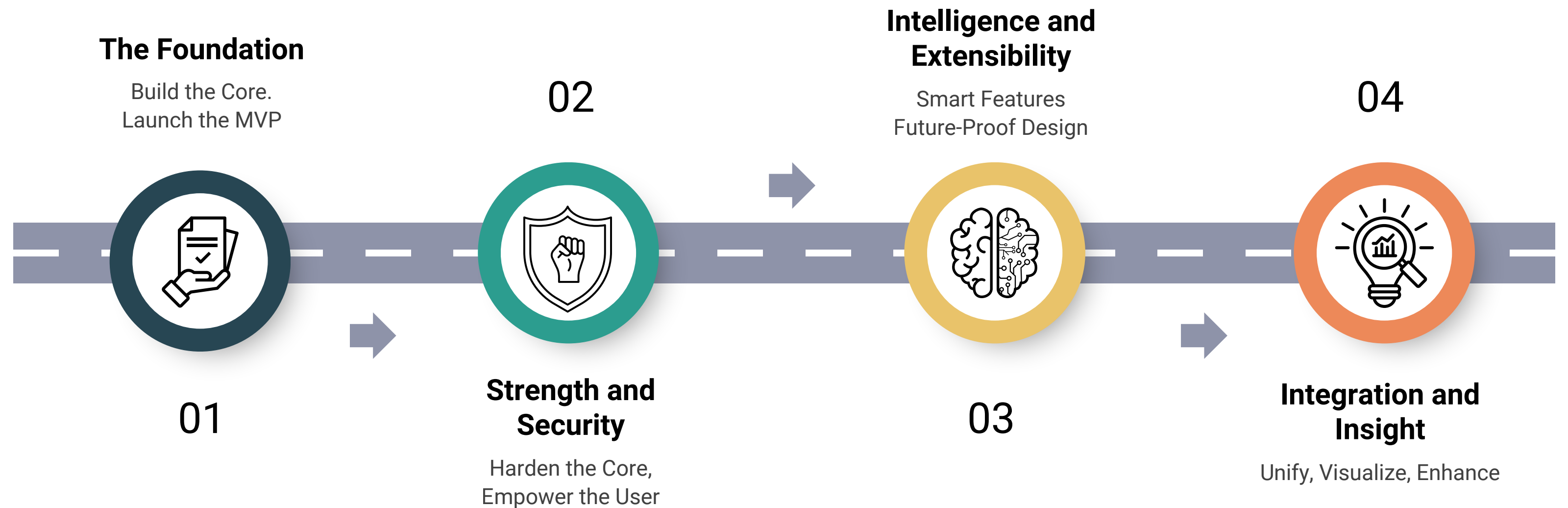


Progress Since FYP Proposal

Development Philosophy

Agile. Modular. User-Centric.

- **Goal:** A functional MVP after each iteration.
- **Testing:** Project testing within and after every iteration

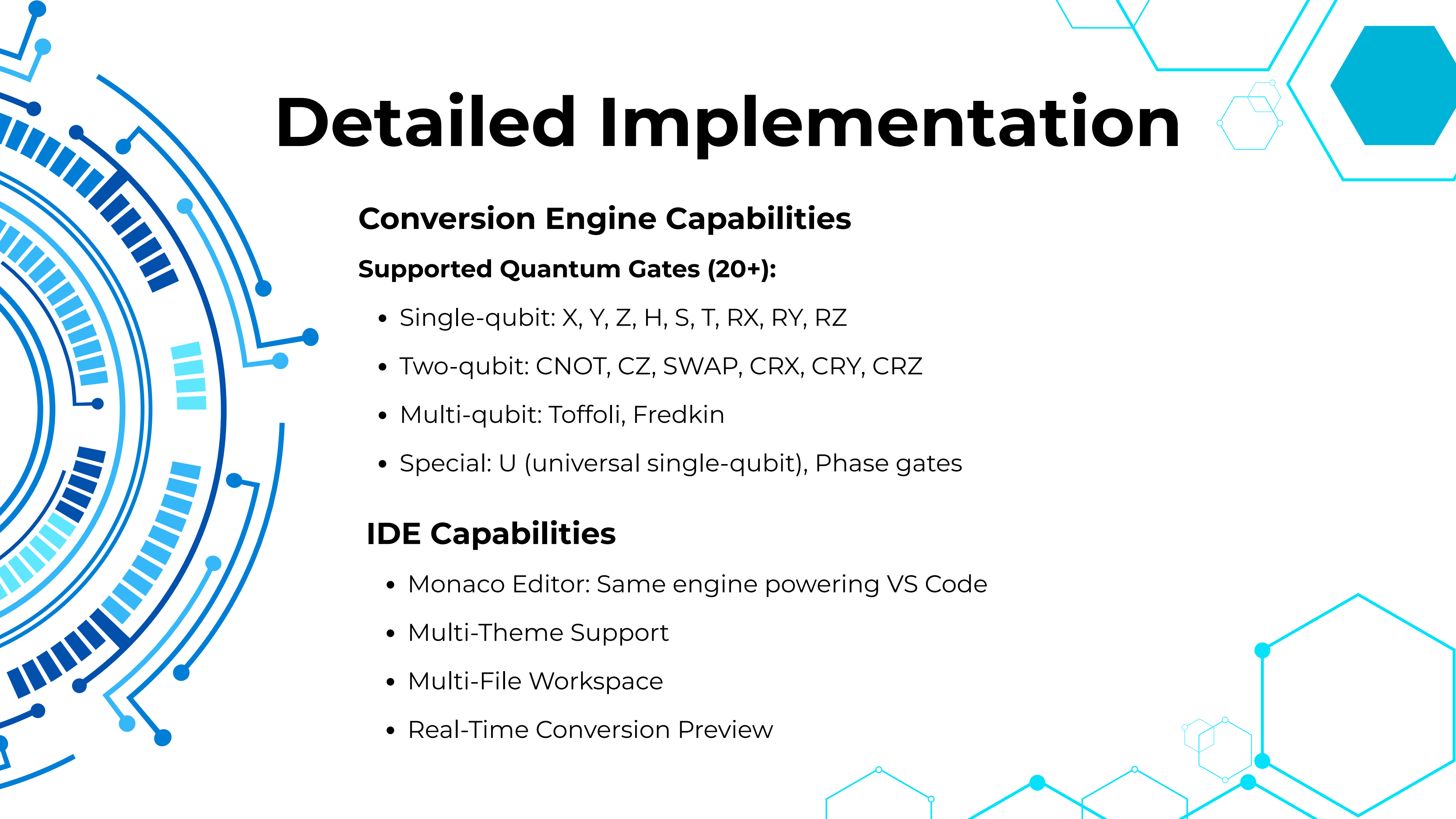




Sprint 1: The Foundation

Fully Completed Sprint-1

- 
- Web IDE
 - Backend API Foundation
 - **Dual-path strategy:** AST-first with safe runtime execution fallback
 - **Compilation Engine:** Gate mapping library for 20+ quantum gates
 - **Multi-Language Support:** Qiskit, Cirq, PennyLane.
 - **Deliverable:** A working, unified compiler and interface.



Detailed Implementation

Conversion Engine Capabilities

Supported Quantum Gates (20+):

- Single-qubit: X, Y, Z, H, S, T, RX, RY, RZ
- Two-qubit: CNOT, CZ, SWAP, CRX, CRY, CRZ
- Multi-qubit: Toffoli, Fredkin
- Special: U (universal single-qubit), Phase gates

IDE Capabilities

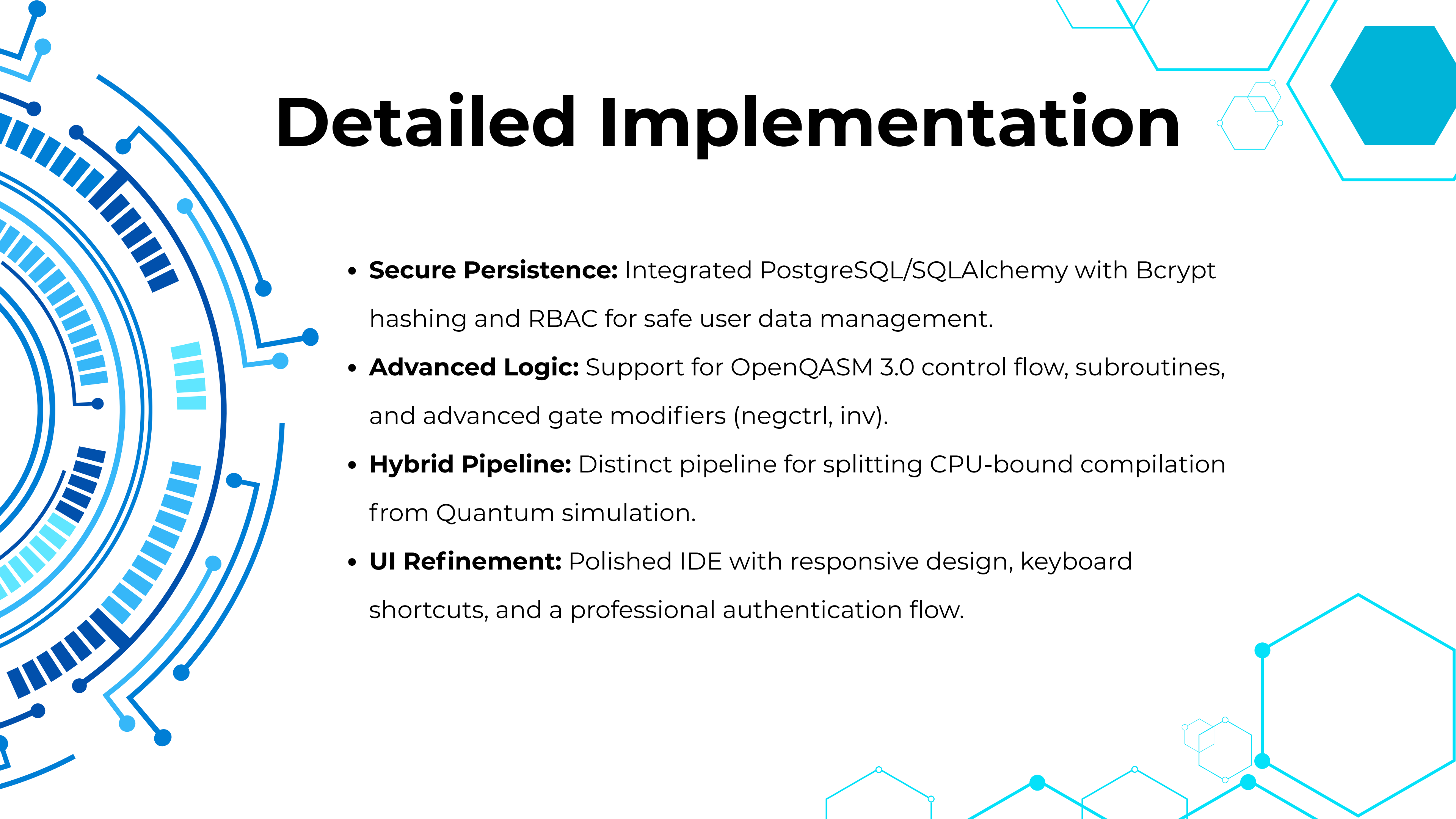
- Monaco Editor: Same engine powering VS Code
- Multi-Theme Support
- Multi-File Workspace
- Real-Time Conversion Preview



Sprint 2: Strength & Security

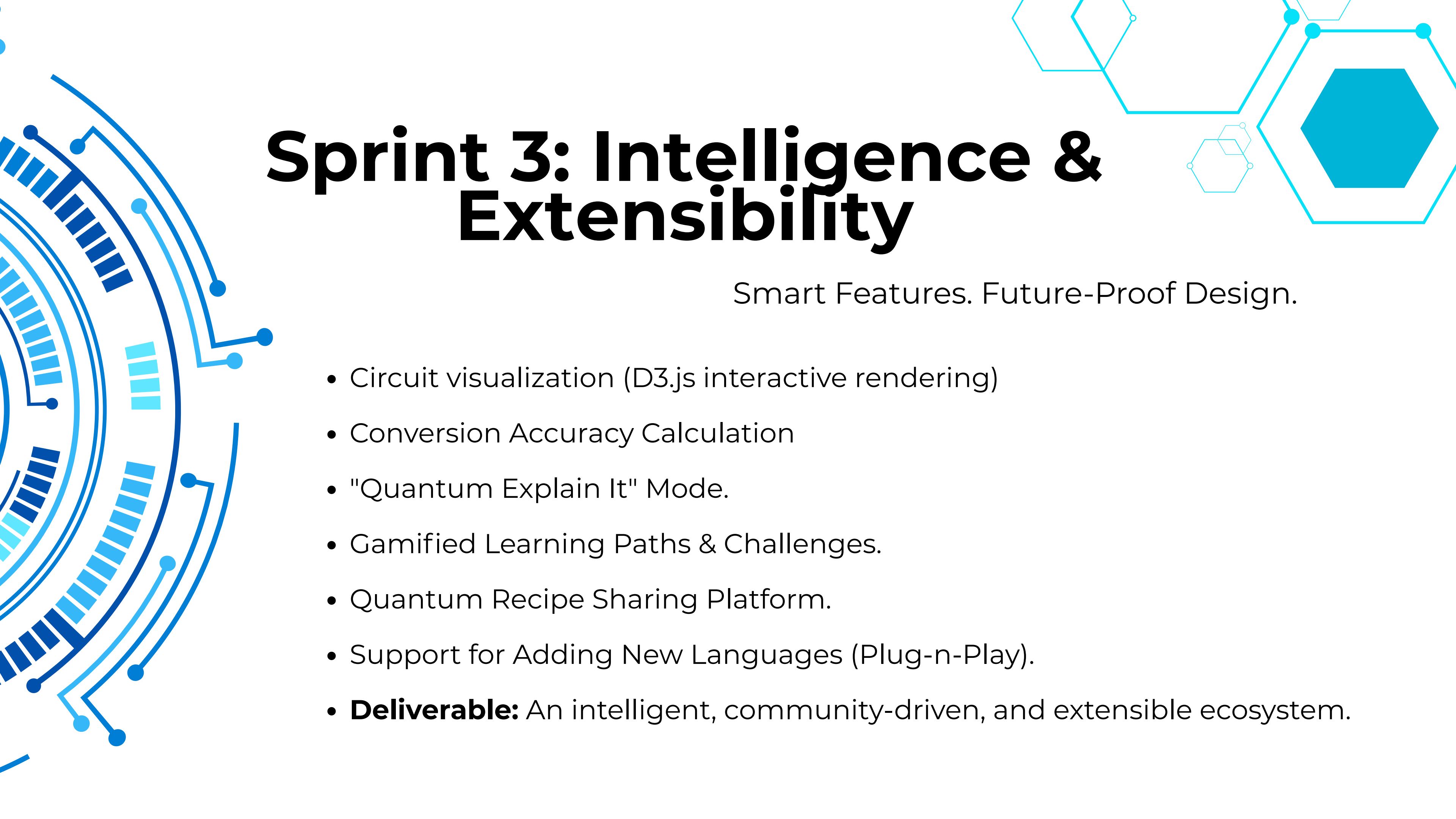
Fully Completed Sprint-2

- **Security & Persistence:** CIA Compliance and Database Layer.
- **Advanced Quantum Features:** Extended Language Support.
- **Hybrid Architecture:** CPU/QPU Orchestration.
- **Refinement:** Benchmarking and UI Polish.
- **Deliverable:** A secure, scalable, and more powerful platform.



Detailed Implementation

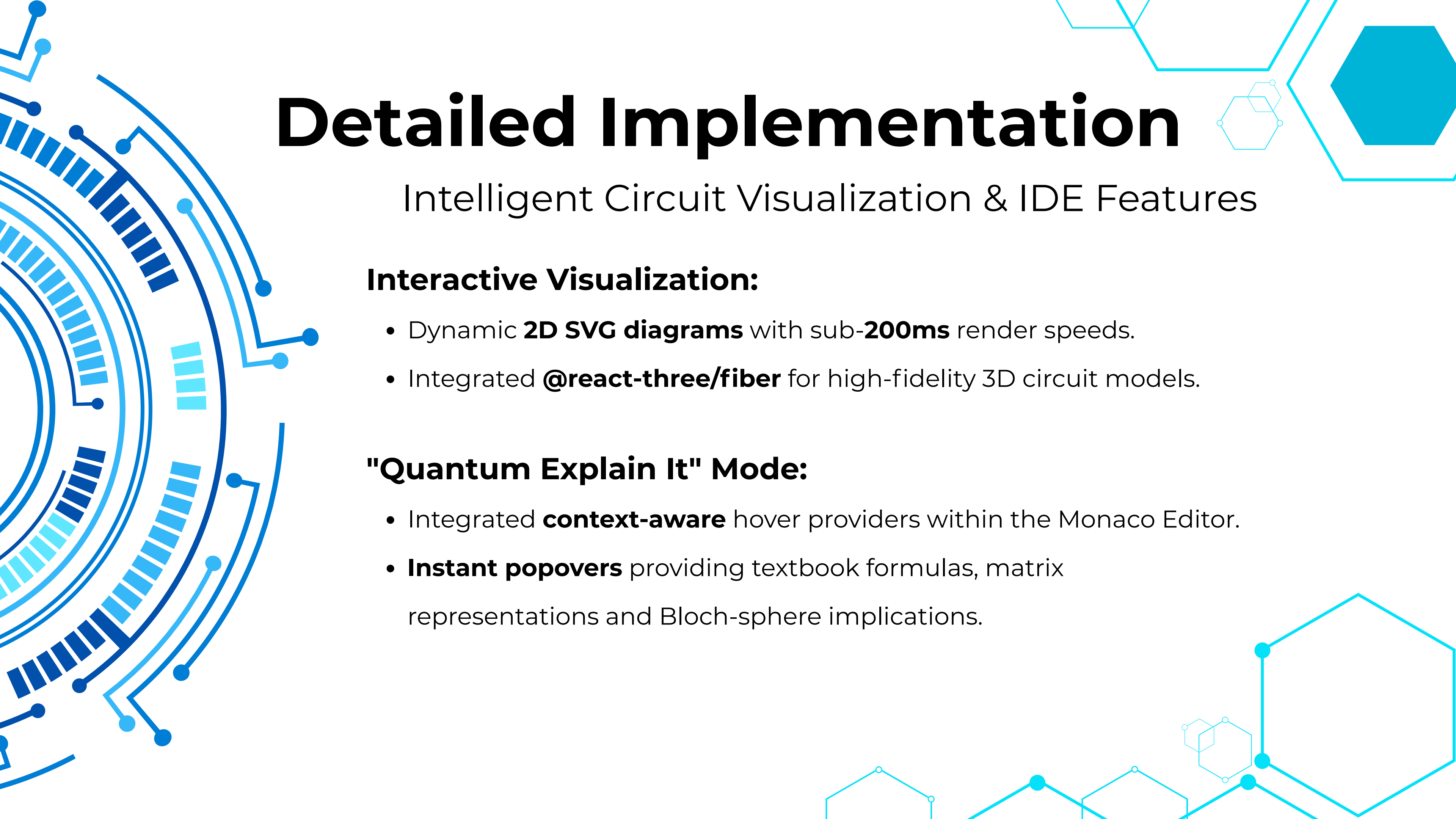
- **Secure Persistence:** Integrated PostgreSQL/SQLAlchemy with Bcrypt hashing and RBAC for safe user data management.
- **Advanced Logic:** Support for OpenQASM 3.0 control flow, subroutines, and advanced gate modifiers (negctrl, inv).
- **Hybrid Pipeline:** Distinct pipeline for splitting CPU-bound compilation from Quantum simulation.
- **UI Refinement:** Polished IDE with responsive design, keyboard shortcuts, and a professional authentication flow.



Sprint 3: Intelligence & Extensibility

Smart Features. Future-Proof Design.

- Circuit visualization (D3.js interactive rendering)
- Conversion Accuracy Calculation
- "Quantum Explain It" Mode.
- Gamified Learning Paths & Challenges.
- Quantum Recipe Sharing Platform.
- Support for Adding New Languages (Plug-n-Play).
- **Deliverable:** An intelligent, community-driven, and extensible ecosystem.



Detailed Implementation

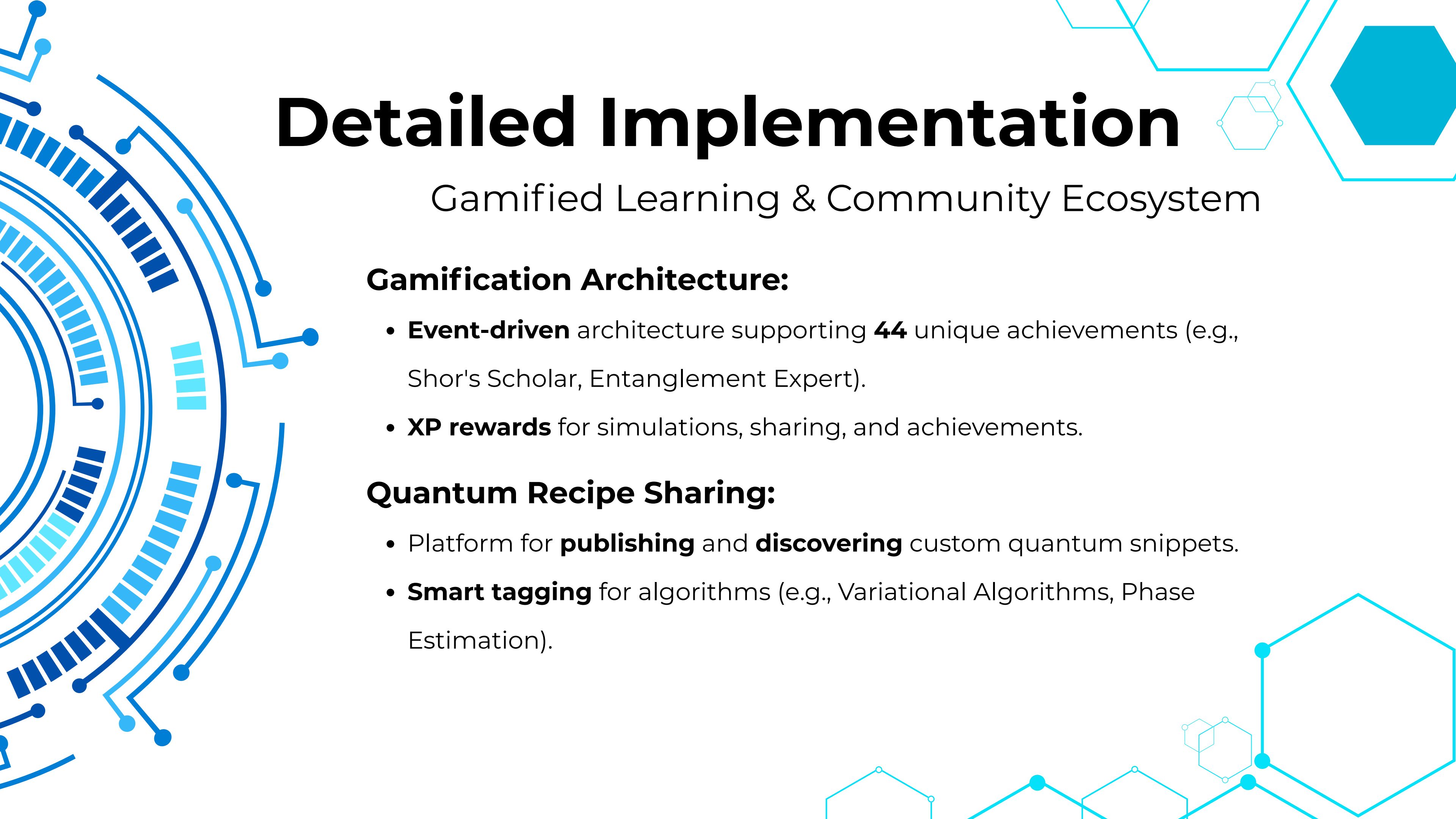
Intelligent Circuit Visualization & IDE Features

Interactive Visualization:

- Dynamic **2D SVG diagrams** with sub-**200ms** render speeds.
- Integrated **@react-three/fiber** for high-fidelity 3D circuit models.

"Quantum Explain It" Mode:

- Integrated **context-aware** hover providers within the Monaco Editor.
- **Instant popovers** providing textbook formulas, matrix representations and Bloch-sphere implications.



Detailed Implementation

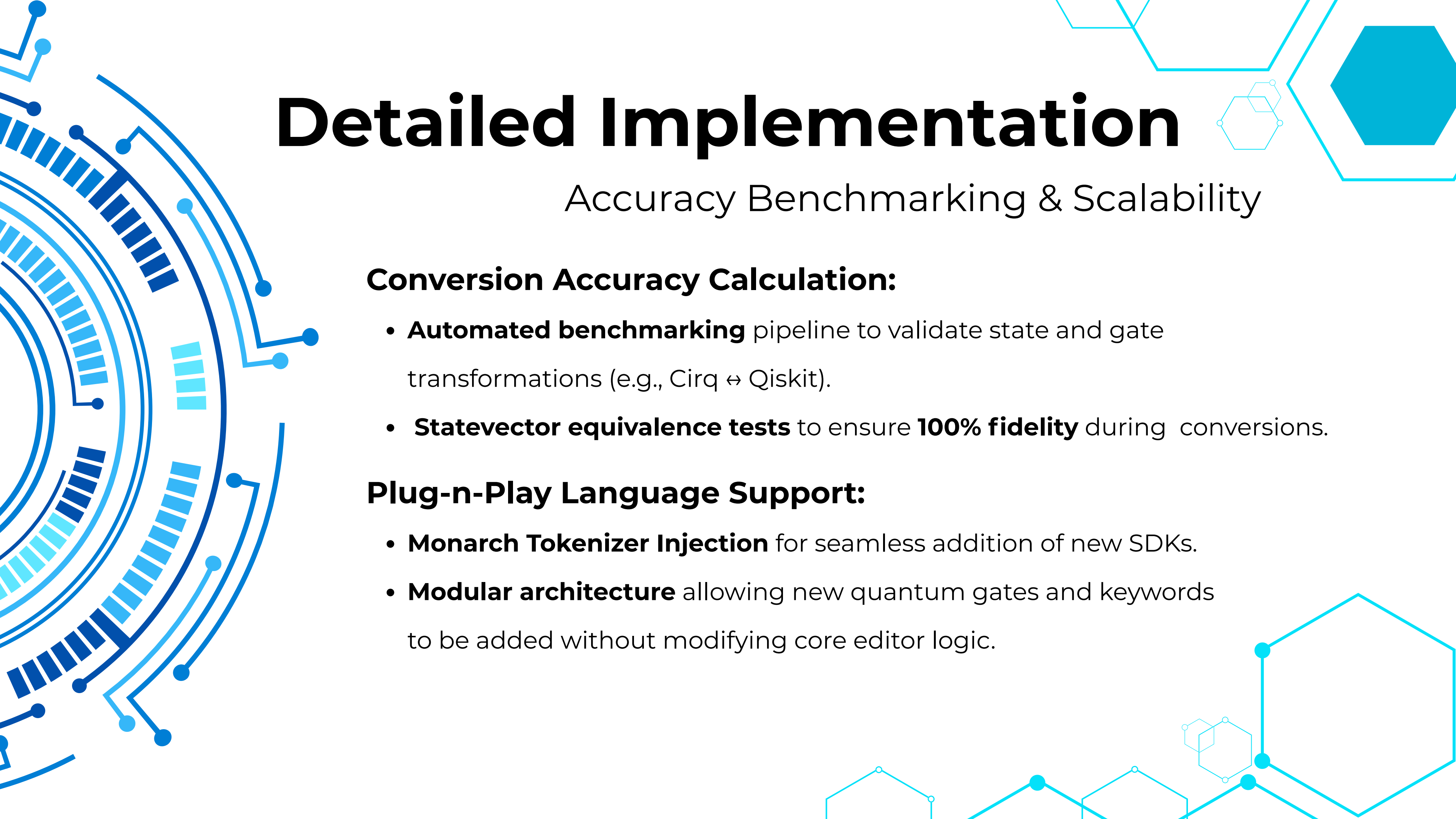
Gamified Learning & Community Ecosystem

Gamification Architecture:

- **Event-driven** architecture supporting **44** unique achievements (e.g., Shor's Scholar, Entanglement Expert).
- **XP rewards** for simulations, sharing, and achievements.

Quantum Recipe Sharing:

- Platform for **publishing** and **discovering** custom quantum snippets.
- **Smart tagging** for algorithms (e.g., Variational Algorithms, Phase Estimation).



Detailed Implementation

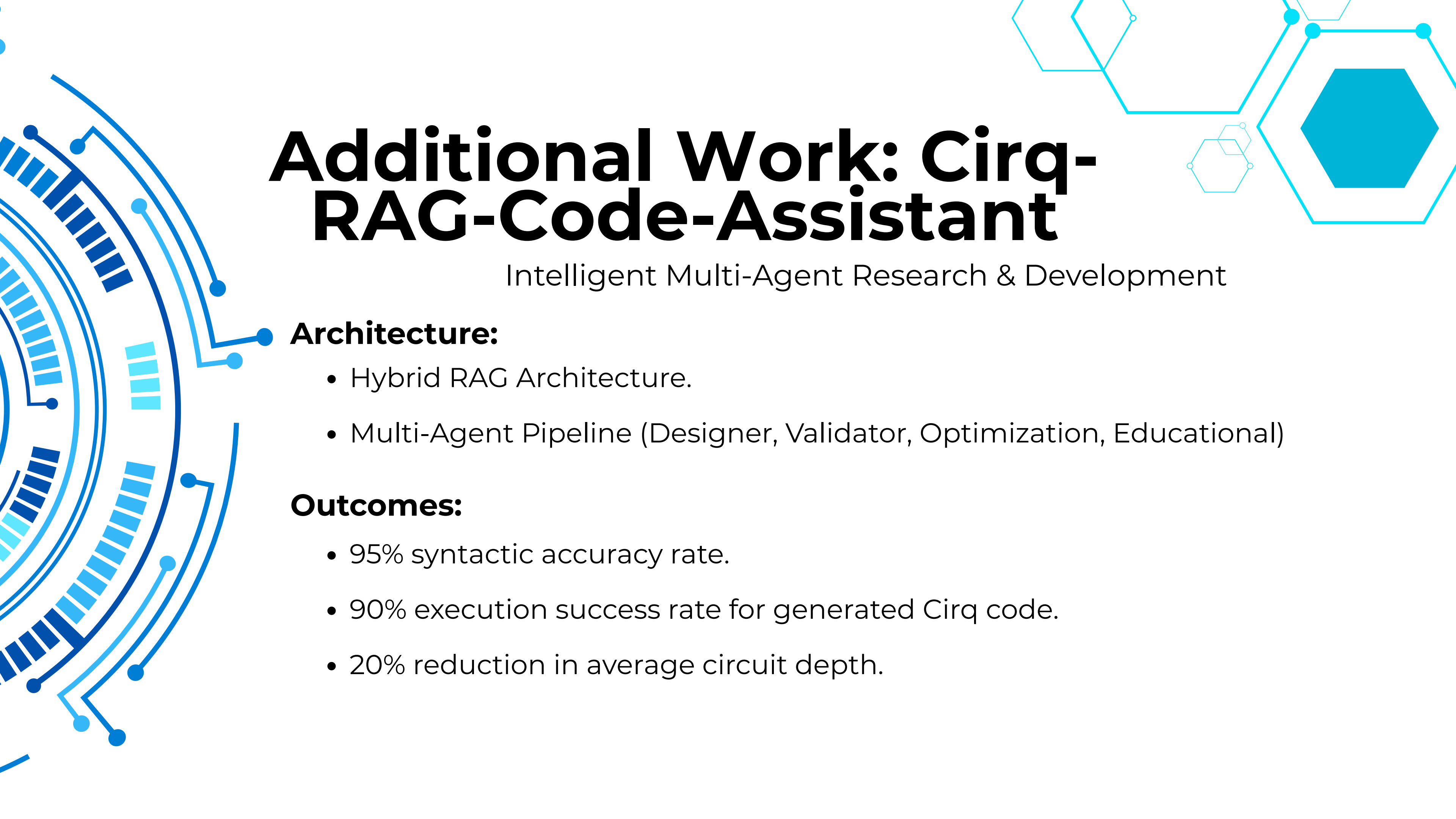
Accuracy Benchmarking & Scalability

Conversion Accuracy Calculation:

- **Automated benchmarking** pipeline to validate state and gate transformations (e.g., Cirq $\leftrightarrow$ Qiskit).
- **Statevector equivalence tests** to ensure **100% fidelity** during conversions.

Plug-n-Play Language Support:

- **Monarch Tokenizer Injection** for seamless addition of new SDKs.
- **Modular architecture** allowing new quantum gates and keywords to be added without modifying core editor logic.



Additional Work: Cirq-RAG-Code-Assistant

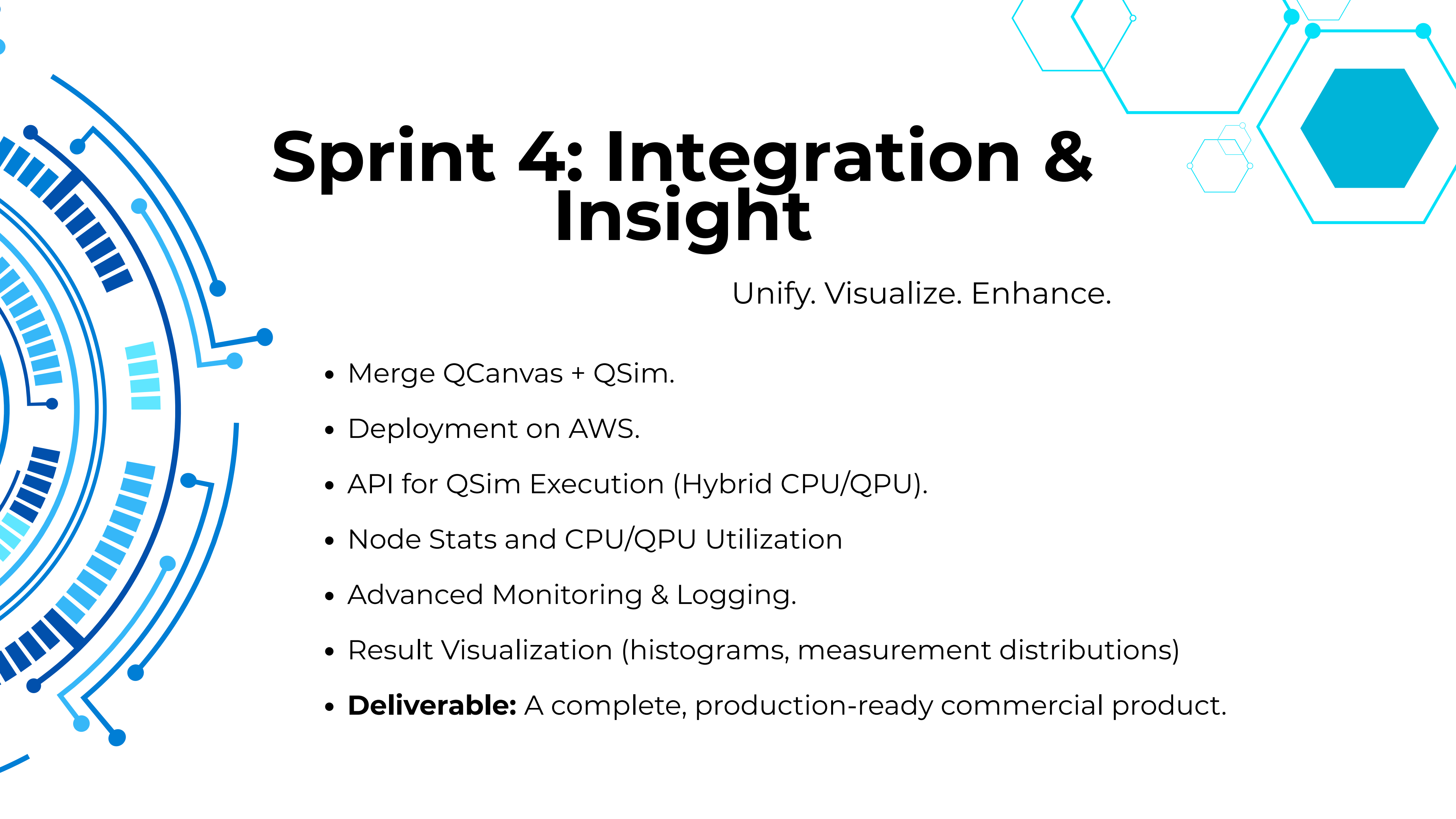
Intelligent Multi-Agent Research & Development

Architecture:

- Hybrid RAG Architecture.
- Multi-Agent Pipeline (Designer, Validator, Optimization, Educational)

Outcomes:

- 95% syntactic accuracy rate.
- 90% execution success rate for generated Cirq code.
- 20% reduction in average circuit depth.



Sprint 4: Integration & Insight

Unify. Visualize. Enhance.

- Merge QCanvas + QSim.
- Deployment on AWS.
- API for QSim Execution (Hybrid CPU/QPU).
- Node Stats and CPU/QPU Utilization
- Advanced Monitoring & Logging.
- Result Visualization (histograms, measurement distributions)
- **Deliverable:** A complete, production-ready commercial product.



Hands-On Quantum Workshop (IIUI)



Workshop Overview

From Bits to Qubits

The Mission:

- A **hands-on** workshop
- Designed for **absolute beginners**
- Transition from theory to **practical** execution.

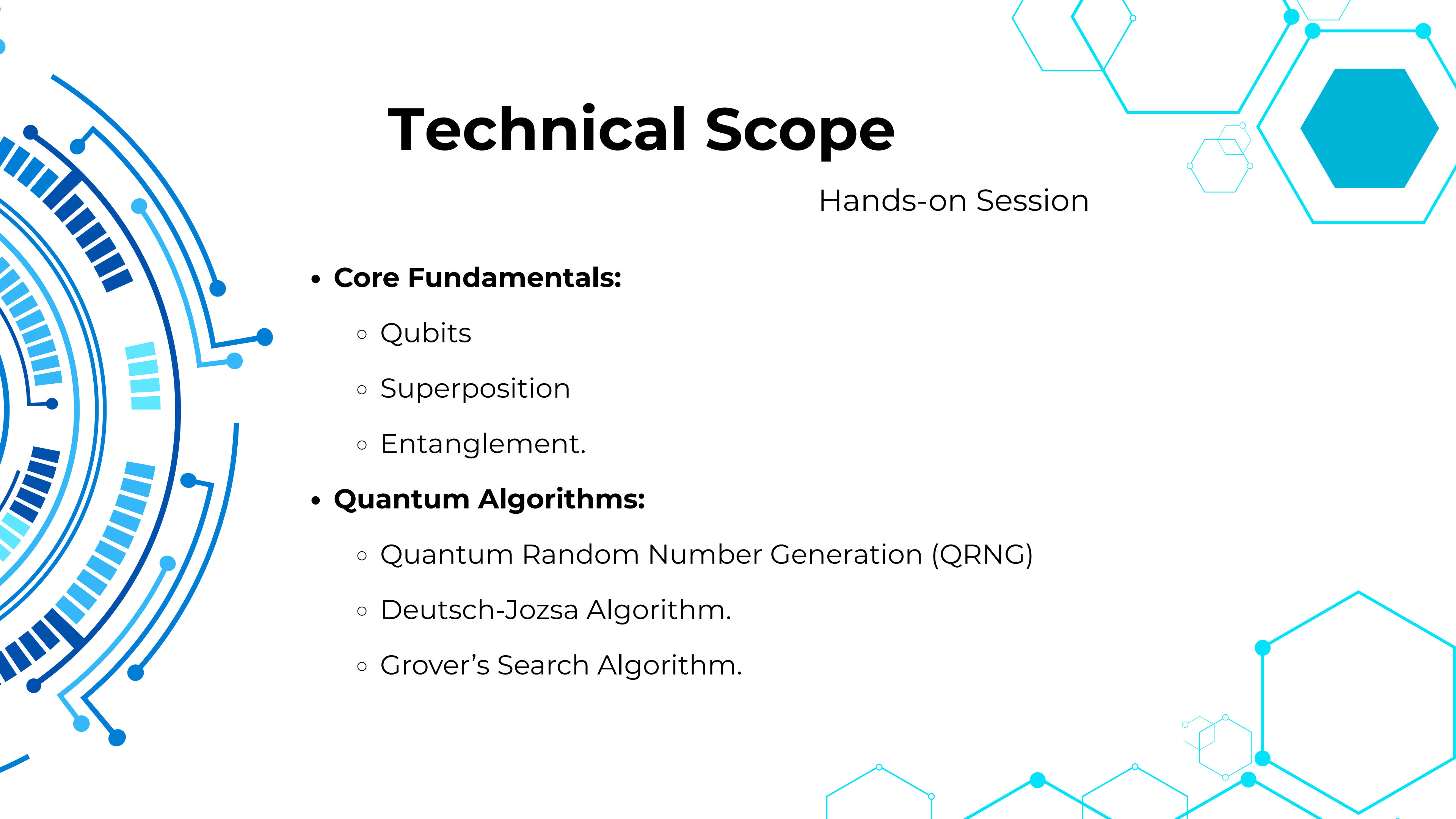
The Organizers:

- QCanvas Team
- QSim Team

Resource Person

- Dr. Imran Ashraf

Date: Dec 29, 2025



Technical Scope

Hands-on Session

- **Core Fundamentals:**

- Qubits
- Superposition
- Entanglement.

- **Quantum Algorithms:**

- Quantum Random Number Generation (QRNG)
- Deutsch-Jozsa Algorithm.
- Grover's Search Algorithm.

Outcome and Impact

Total Participants: 53

Experience Levels:

- 65% Beginners
- 30% Basic Knowledge
- 5% Intermediate

Proof of Concept:

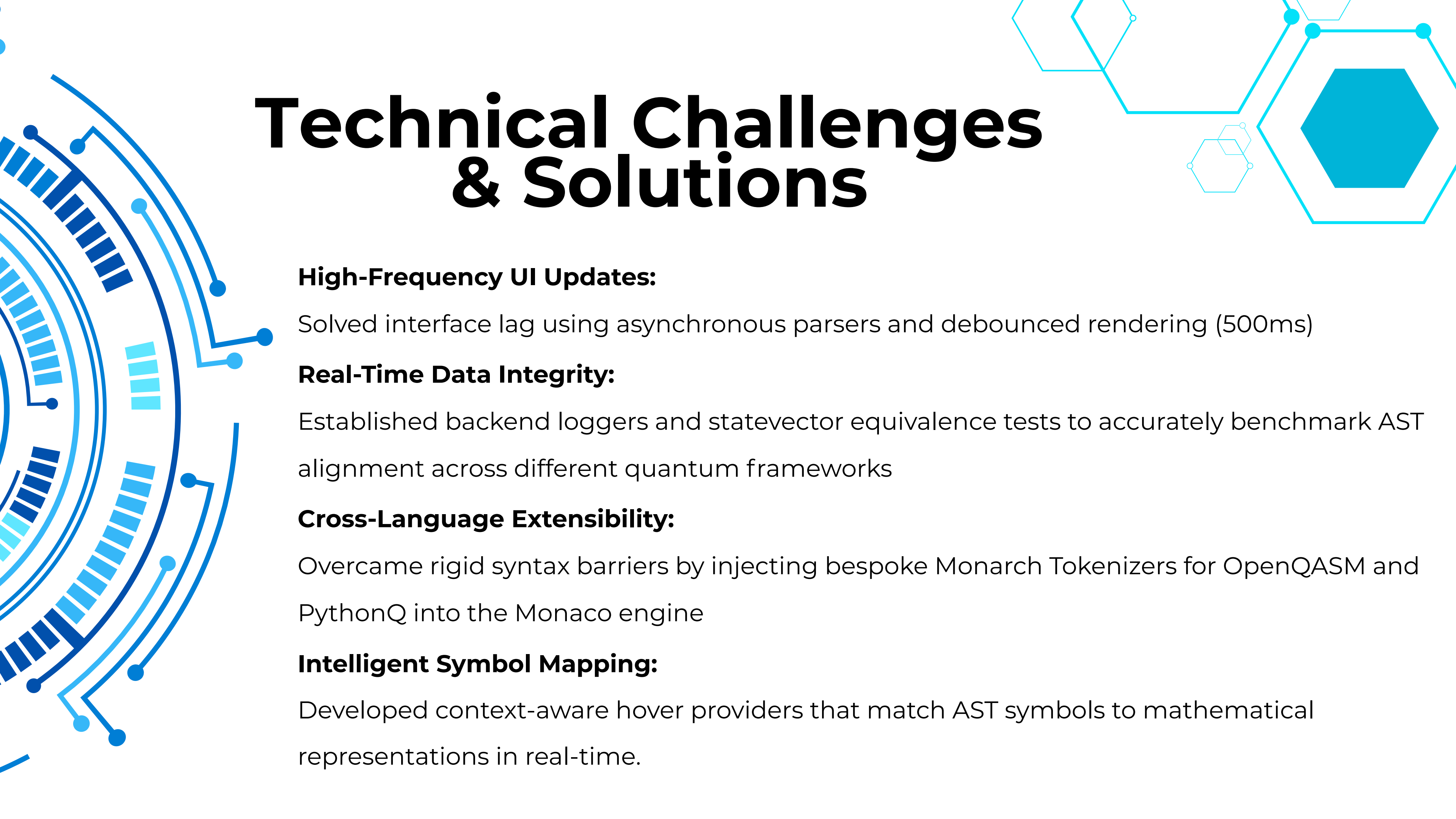
Demonstrated strong demand for advanced computing topics.

Brand Value:

Enhanced FAST's reputation as an innovator in tech education.



Technical Challenges & Solutions



Technical Challenges & Solutions

High-Frequency UI Updates:

Solved interface lag using asynchronous parsers and debounced rendering (500ms)

Real-Time Data Integrity:

Established backend loggers and statevector equivalence tests to accurately benchmark AST alignment across different quantum frameworks

Cross-Language Extensibility:

Overcame rigid syntax barriers by injecting bespoke Monarch Tokenizers for OpenQASM and PythonQ into the Monaco engine

Intelligent Symbol Mapping:

Developed context-aware hover providers that match AST symbols to mathematical representations in real-time.



Testing & Validation Strategy



Testing Framework



Test Coverage:

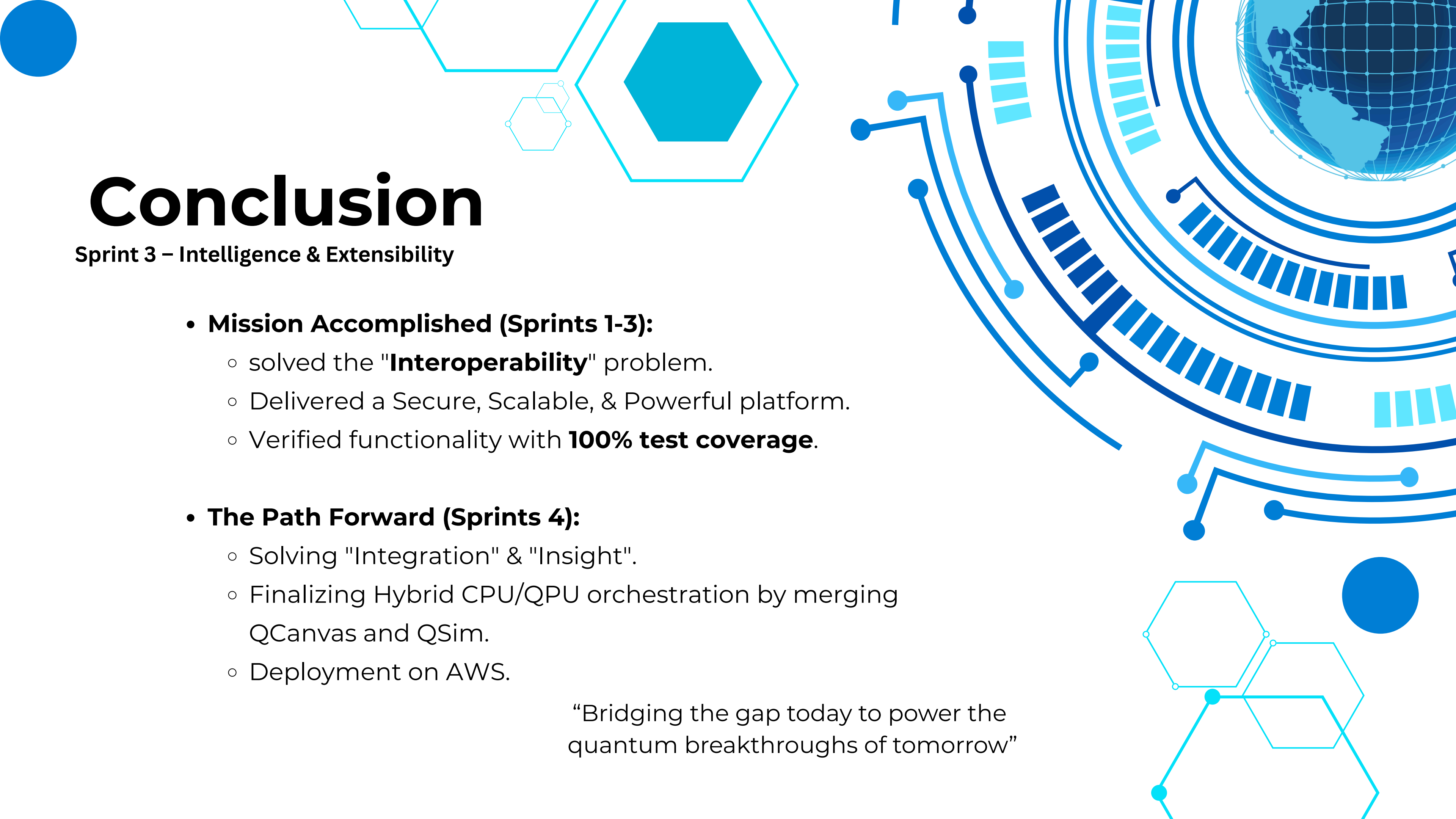
- **2,359** functional and unit tests.
- Test suite scaled by **7.7x** since Sprint 2.
- **Framework Support:** Full validation for Cirq, Qiskit, PennyLane, OpenQASM 3.0, and custom PythonQ wrappers.

Validation Results:

- **100% fidelity** in AST-based conversions between frameworks.
- **95%** syntactically correct code generation.
- Automated error detection and context-aware debugging for **20+** simultaneous qubits.



Conclusion



Conclusion

Sprint 3 – Intelligence & Extensibility

- **Mission Accomplished (Sprints 1-3):**
 - solved the "**Interoperability**" problem.
 - Delivered a Secure, Scalable, & Powerful platform.
 - Verified functionality with **100% test coverage**.
- **The Path Forward (Sprints 4):**
 - Solving "Integration" & "Insight".
 - Finalizing Hybrid CPU/QPU orchestration by merging QCanvas and QSim.
 - Deployment on AWS.

“Bridging the gap today to power the quantum breakthroughs of tomorrow”

Any Questions ?

